UNIVERSITY OF MORATUWA

Faculty of Engineering

Registered Module No: ME3993

INDUSTRIAL TRAINING REPORT

Yugadanavi 300MW Combined Cycle Power Plant

From: 27/11/2023 To 30/05/2024

Date of Submission:

06/06/2024

A. G. C. O. Atadaswala

200055B

Department of Mechanical Engineering

PREFACE

This document presents the experience and knowledge I gathered during my 26-week industrial training at the YUGADANAVI Combined Cycle Power Plant in Kerawalapitiya. During this period, I had the opportunity to gain practical experience and apply my theoretical knowledge as a mechanical engineer. The training took place from November 27, 2023, to May 30, 2024, at the YUGADANAVI Power Plant, the first 300 MW combined cycle power plant in Sri Lanka.

Throughout the training, I focused on enhancing my technical skills, management abilities, communication proficiency, leadership qualities, problem-solving aptitude, decision-making skills, and time management. Moreover, I learned about the professional ethics essential in the industry, which enabled me to develop a strong foundation for my future career.

This report aims to provide a detailed overview of my experiences and the exposure I gained during the training period. It is mainly divided into three chapters, each focusing on different aspects of my training.

The first chapter provides background information about the power plant, including its mission, vision, organizational structure, and a SWOT analysis to assess its strengths, weaknesses, opportunities, and threats.

In the second chapter, I will describe the specific areas I was exposed to during my training, including the projects I worked on, the tasks I was assigned, and my involvement in administrative and office practices. I will also share my hands-on experience with machinery and equipment.

The final chapter offers a summary of my training program and how it aligned with the intended learning outcomes. Additionally, I will provide suggestions and proposed improvements that were presented to the company to enhance the effectiveness of work processes, mainly in the engineering context at YUGADANAVI CCPP.

ACKNOWLEDGMENT

The industrial training period was a significant opportunity for me to enhance my technical skills, communication abilities, and overall confidence in overcoming challenges. However, this achievement would not have been possible without the support and guidance of numerous individuals. I would like to express my sincere gratitude to all those who played a part in my successful completion of the industrial training program.

First and foremost, I would like to extend my thanks to Eng. Roy Sankaranarayana, the Director of the Industrial Training Division at the University of Moratuwa, and Prof. S. Witharana and Dr. Lihil Uthpala Subasinghe, the department coordinators of industrial training for Mechanical Engineering. Their guidance and support throughout the training period were invaluable. I am also grateful to all the members of the Industrial Training Division at the Faculty of Engineering, University of Moratuwa, and NAITA (National Apprentice and Industrial Training Authority) for providing us with this wonderful opportunity.

I would like to express my heartfelt appreciation to the top management of the Engineering Department of YUGADANAVI Combined Cycle Power Plant. Special thanks go to Mr. Thilina Ranasinghe, the Senior Mechanical Engineer, and Mr. Nadun Wijesinghe, a Mechanical Engineer and my supervisor, for guiding me and teaching me throughout the training program. I also want to thank Mr. Kamal Peiris, EHS Manager and training coordinator at YUGADANAVI Power Plant. I am also grateful to all the mechanical engineers, electrical engineers, assistant engineers, and mechanical and electrical superintendents and technicians at YUGADANAVI CCPP for their assistance and guidance throughout my time at the establishment.

Finally, I would like to extend my gratitude to all the individuals who supported me during my internship at YUGADANAVI CCPP. Although not mentioned by name, their contributions were critical in making my training experience a success.

A.G.C.O.Atadaswala,

200055B,

Engineering undergraduate,

Department of Mechanical Engineering,

University of Moratuwa.

Contents

List of Figures

List of Tables

Abbreviations

- CCPP – Combined cycle power plant
- HRSG – Heat recovery steam generator
- WTP – Water treatment plant
- HFO – Heavy fuel oil
- FOTP – Fuel oil treatment plant
- DO – Diesel oil
- CCW – Closed cooled water
- SWRO – Sea water reverse osmosis
- BWRO – Brackish water reverse osmosis
- CRP- Condensate recirculation pump
- GT – Gas turbine
- LNG – Liquefied natural gas
- ST – Steam turbine
- LP – Low pressure
- HP – High pressure
- DM – Demineralized
- ETP – Effluent treatment plant
- CT – Cooling Tower
- CI – Combustion inspection
- MI – Main inspection
- PPE – Personal Protective Equipment
- ACW – Auxiliary cooling water
- HGPI – Hot gas path inspection
- BFP – Boiler feed water pump
- UF – Ultra filtration

1 INTRODUCTION TO TRAINING ESTABLISHMENT

1.1 Introduction to Lakdhanavi Ltd

West Coast Power (Pvt) Ltd is a vital arm of Lakdhanavi Limited, a flagship subsidiary within the distinguished LTL Holdings. Since the 1980s, LTL Holdings has become the leading engineering group in Sri Lanka, extending its influence globally across five countries: Sri Lanka, Bangladesh, India, Nepal, and Singapore. Renowned for its commitment to engineering excellence, LTL Holdings specializes in the production of transformers and switchgear, distributed across more than 25 countries.

Yugadanavi Power Plant, a 300 MW combined cycle power plant, is established at Kerawalapitiya. Lakdhanavi Limited, a subsidiary of LTL Holdings (Pvt) Ltd, has undertaken the operations and maintenance of the Yugadanavi Power Plant. As a global leader in the power sector, Lakdhanavi stands as a pioneer in Independent Power Production (IPP), Engineering, Procurement, and Construction (EPC) contracting, and Operations and Maintenance (O&M) contracting for thermal and renewable power plants. The company's influence extends beyond Sri Lanka, encompassing markets in Bangladesh, Nepal, the Maldives, and Oman.

Lakdhanavi constructed its first power plant in 1997 and has since significantly contributed to the power sector, involving the largest power generation projects as an Independent Power Producer. They have also received Gold Awards from Asia Power Magazine and were recognized as the Best Engineering Organization at the 2008 Engineering Excellence Awards ceremony organized by the Institution of Engineers Sri Lanka.

In 2007, Lakdhanavi undertook the largest power plant project in Sri Lanka: a 300 MW multi-fuel combined cycle power plant. The completion of this power plant was essential to meet the power demand of the country at that time. This power plant contributes over 22% of the country's energy demand, and together with other power plants operated by Lakdhanavi Ltd, it contributes approximately 35% of the total energy demand of the country.

Moreover, the "Sobadhanavi Limited" project, a 350 MW combined cycle power plant in Kerawalapitiya, Sri Lanka, is a testament to the company's forward-looking approach, as it aims to be Sri Lanka's first LNG-fired power plant. This project not only contributes to the national energy grid but also underscores Lakdhanavi's commitment to embracing cleaner and more sustainable energy sources.

Figure 1 Logo of Lakdhanavi Limited

Figure 2 Logo of LTL Holdings

1.2 Introduction to 300MW YUGADANAVI CCPP

The operations and maintenance of the YUGADANAVI Power Plant have been undertaken by Lakdhanavi Ltd. The configuration of this power plant is 2-2-1, meaning it has two gas turbines, two heat recovery steam generators, and one steam turbine. The plant was commissioned employing local engineers, skilled technicians, and laborers. The Yugadanavi Power Plant is operated and maintained entirely by Sri Lankan personnel. It follows rigorous safety procedures and provides a safe and clean environment for all employees.

The power plant has been constructed near the sea to facilitate the drawing of seawater for plant usage. The steam turbine, cooling towers, and other auxiliary systems consume seawater and purified water derived from seawater. The two gas turbines consume heavy fuel oil (HFO), while the steam turbine runs using demineralized water. The plant is also located near its HFO supplier, Ceylon Petroleum Storage Terminal Limited (CPSTL).

The plant can be divided into three main sections: operation, maintenance, and safety. The operation section is the main section of the power plant, where all systems are monitored and controlled. All maintenance tasks are handled by the maintenance section. Safety procedures are overseen by the safety manager. All employees are required to be familiar with safety procedures to ensure human protection. Training programs and fire drills are conducted at the power plant to prepare for any accidents or hazards.

Figure 3 Yugadanavi CCPP

1.2.1 Vision

To become the preferred partner in the global power sector by offering unparalleled expertise and cost-efficient solutions, while aim is to create a sustainable future through delivering excellence in all endeavours.

1.2.2 Mission

Our mission is to leverage our knowledge and expertise in independent power production, engineering, procurement and construction, and the operation and maintenance of power plants. We are committed to enhancing the quality of life for people worldwide.

1.2.3 SWOT analysis

Strengths

The combined cycle technology used in the plant is highly efficient, harnessing both gas and steam turbines to maximize electricity generation from the same amount of fuel. This technology ensures a stable and reliable power supply, which is essential for meeting Sri Lanka's energy demands. The plant is known for its high-quality service and can start up more quickly from standby mode than other plants in the country, providing electricity to the CEB. It operates 24/7 with over 15 years of experience with skilled engineers and technicians ensuring smooth operations and maintenance. Its strategic location near Colombo allows easy access to seawater and enables efficient power distribution to the most industrialized and populous region of Sri Lanka.

Weaknesses

The plant primarily uses Heavy Fuel Oil (HFO) and Diesel, making it dependent on fuel imports and vulnerable to global fuel price fluctuations. Its advanced technology and components demand specialized skills for operations and maintenance, including tasks like HGPI, CI, and MI, which contribute to higher operational costs. While it is more efficient than traditional power plants, it still generates greenhouse gas emissions and pollutants, including sulphur-containing gases.

Opportunities

Sri Lanka's growing energy demand presents numerous opportunities for expanding capacity and enhancing the plant's performance. Building and developing new power plants will increase electricity supply. There's also potential to receive government support and incentives to improve and make the infrastructure more efficient.

Threads

The growing use of renewable energy sources like solar presents a significant challenge, as it has driven down costs. Additionally, fluctuations in the availability and price of HFO fuel can be problematic, influenced by political events, high inflation in Sri Lanka, or other disruptions, potentially impacting the plant's operations.

1.2.4 Organizational Structure

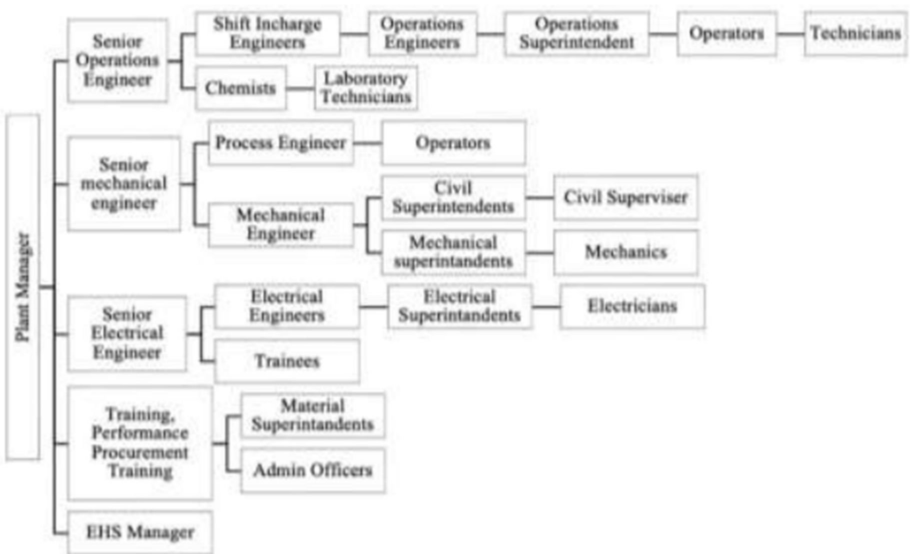

Figure 4 Organizational Structure

2 TRAINING EXPOSURE AND EXPERIENCE

2.1 Working Environment

2.1.1 300MW Combined Cycle Power Plant

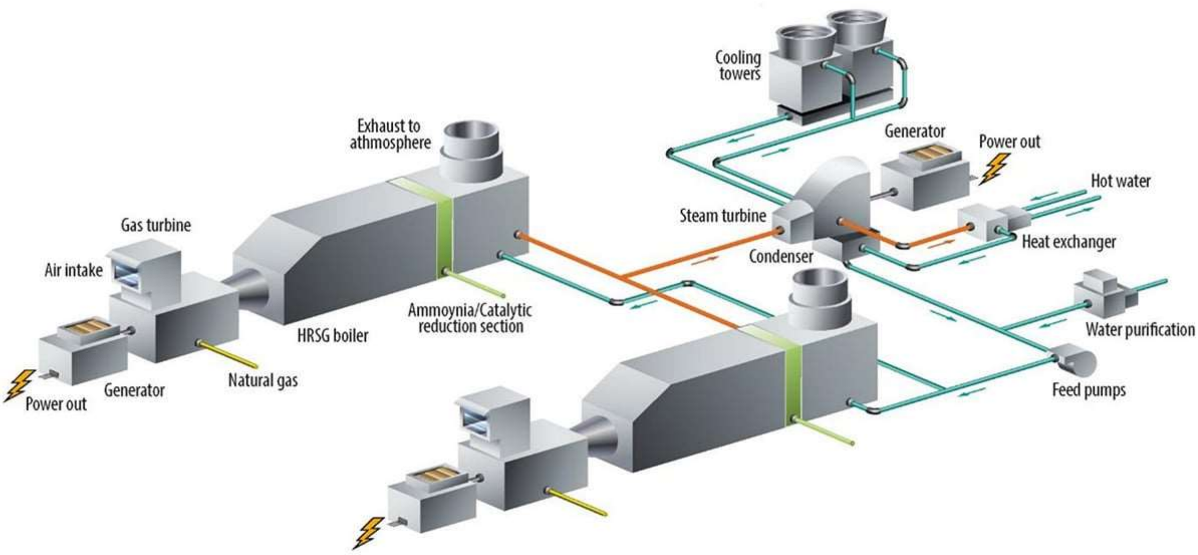

Figure 5 Structure of a combined cycle power plant

Operating with a combined cycle configuration, the plant efficiently utilizes two gas turbines, two heat recovery steam generators, and one steam turbine. It can operate on various fuels, primarily Heavy Fuel Oil (HFO) and Diesel, and has the capability to use Liquefied Natural Gas (LNG).

Strategically located near the sea, the plant easily accesses seawater for processes like the operation of the Steam Turbine and Cooling Towers. The two Gas Turbines operate on Heavy Fuel Oil (HFO), while the Steam Turbine uses demineralized water. Close proximity to its HFO supplier, Ceylon Petroleum Storage Terminal Limited (CPSTL), enhances operational efficiency.

The plant is organized into three main sections: Operation, Maintenance, and Safety. The operation section monitors and controls all systems, while Maintenance ensures smooth plant operation. Safety procedures are rigorously managed under a dedicated safety manager, with regular training programs and fire drills ensuring a safe environment for employees.

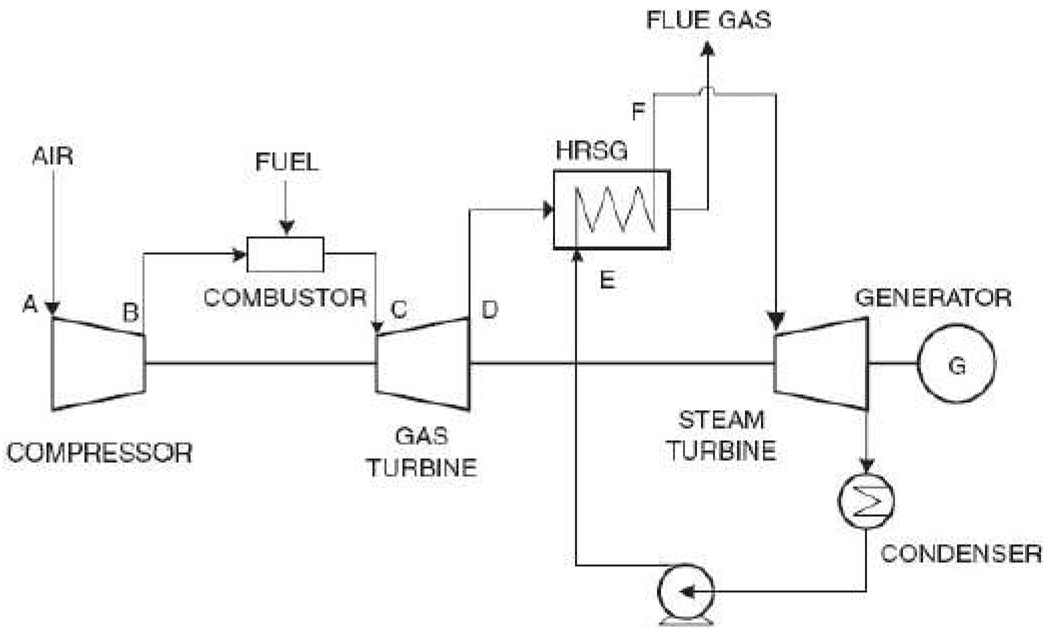

Figure 6 Cycles in CCPP

2.1.1.1 Gas Turbine

Theory

The thermodynamics cycle upon which all gas turbines operate is called the Brayton cycle.

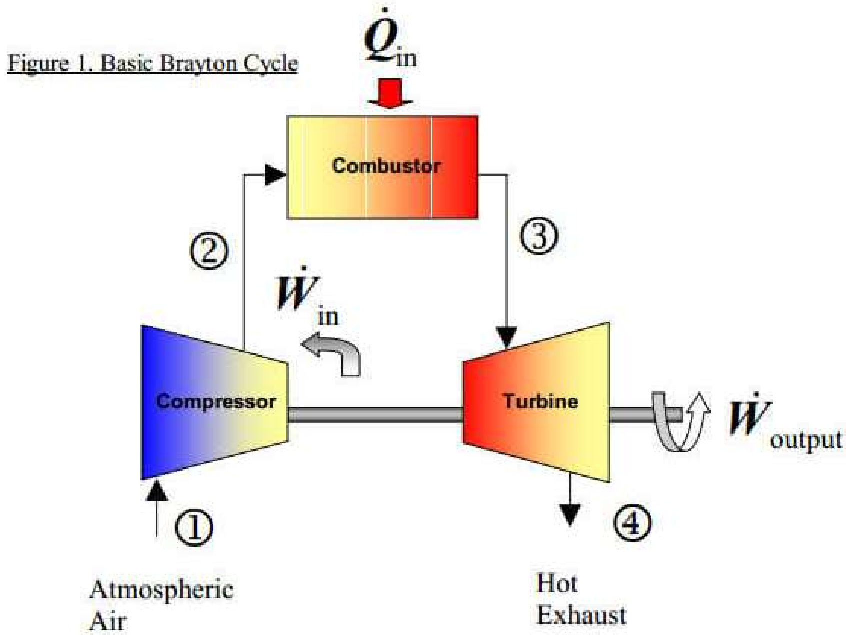

Figure 7 Brayton Cycle

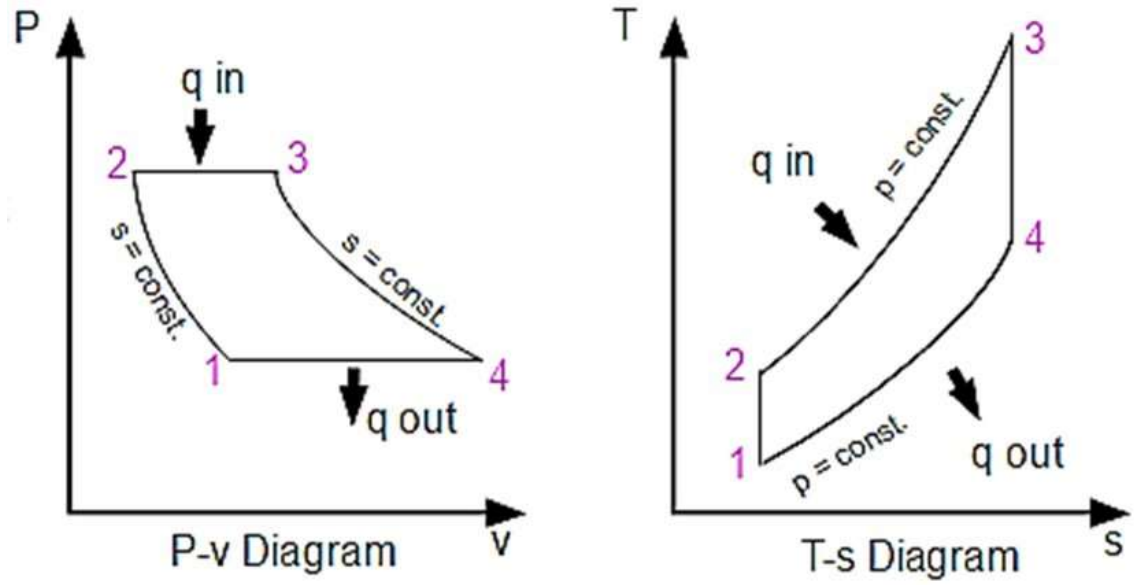

Figure 8 P-V & T-s Diagrams in Brayton cycle

1 to 2: - Isentropic Compression

2 to 3: - Constant pressure heat addition

3 to 4: - Isentropic Expansion

4 to 1: - Constant pressure heat rejection

Gas Turbine Technical Data

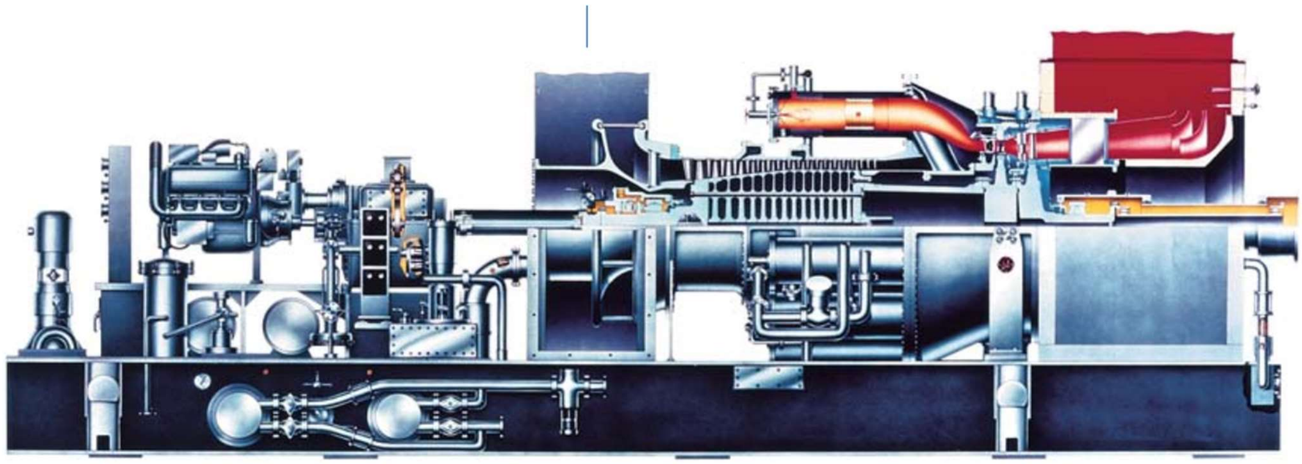

Figure 9 Half cut of the gas turbine.

- Type: GE PG9171 (E), HEAVY FUEL OIL GAS TURBINE
- Brand: GE Energy
- Load Condition: BASE
- Fuel Type: Liquid
- Fuel Temperature: 125 Celsius degrees.
- Output: 102500 kW
- Auxiliary Power: 362 kW
- Net Output: 102138 kW
- Exhaust Flow: 1450.1*103 kg/hr.
- Exhaust Temperature: 512.8 Celsius degrees.
- Exhaust Energy: 749.8 GJ/hr.

This gas turbine is consisting of four major compartments.

1. Auxiliary compartment

The auxiliary compartment of a gas turbine supports startup and ensures optimal operation, housing essential equipment for efficiency and safety. It includes:

- **Turning Gear Motor:** Rotates the rotor at low speed (128 rpm) during standby to prevent thermal gradient-induced rotor bowing.

- **Cranking Motor:** Operates at medium speeds (780 rpm) to adjust turbine speed during startup for optimal conditions.

- **Torque Converter:** Generates torque to initiate turbine rotation during startup.

- **Gear Box:** Links to the main fuel pump via a clutch, vital for power transmission.

- **Main Fuel Pump:** Supplies fuel to combustion chambers at necessary pressure, critical for combustion.

- **Flow Divider:** Distributes fuel equally to 14 combustion chambers, ensuring controlled combustion.

- **Atomizing Air Compressor:** Provides excess air to fuel nozzles, enhancing combustion efficiency.

- **Lube Oil Pumps:** Supplies lubricating oil to turbine bearings for smooth operation, including a mechanical pump coupled with the gearbox, an AC pump for cranking, and an emergency pump.

- **Lube Oil Coolers:** Regulates lubricating oil temperature to prevent overheating.

- **Inlet Guide Vanes (IGV):** Controls compressed air intake at compressor inlet to optimize performance.

- **Fuel Bypass Valve:** Regulates fuel supply to combustion chambers, enhancing control and safety.

2. Turbine compartment

The turbine compartment in a gas turbine is crucial for power generation, comprising three main components: the main axial-flow compressor, combustion chambers with fuel nozzles, and the turbine section.

- **Compressor:** Responsible for pre-compressing air before it enters combustion chambers, the compressor features 17 stages with rotating blades and stationary stator vanes. It raises air pressure up to 10 bars, achieving a compression ratio of 10:5. Blades are securely held by dovetail platforms on the rotor and stator rings, made from corrosion-resistant stainless steel.
- **Combustion Chamber:** Here, compressed air mixes with atomized fuel from 14 combustion cans, each housing a nozzle. Ignition via two spark plugs in cans 13 and 14 initiates combustion, facilitated by 14 crossfire tubes linking chambers. Transition pieces downstream connect to the first turbine stage nozzle, with four flame detectors (linked to cans 4, 5, 10, and 11) ensuring flame presence. Air enters via swirl tips on fuel nozzles, metering and combustion holes in the liner. Atomizing air aids in fuel spray formation.

Figure 12 Liner

Figure 11 Transfer Piece

Figure 10 Fuel Nozzle

- **Turbine:** The turbine section of a gas turbine processes high-temperature, high-pressure gases from combustion to drive turbine blades across three stages. Each stage features nozzles and buckets that increase in size from the first to the third stage, accommodating gas flow expansion. Stationary nozzles direct hot gases against turbine buckets, rotating the turbine shaft connected to the compressor. Diaphragms on second and third stage nozzles prevent air leakage, featuring labyrinth-type seal teeth. Turbine bucket tips interface with annular turbine shrouds to minimize tip clearance leakage, ensuring efficient energy conversion from thermal to mechanical.

Figure 14 Turbine buckets

Figure 13 Shrouds

Figure 15 Nozzles with diaphragms

3. Load compartment

The load compartment serves to isolate the diffuser from the generator compartment, ensuring safety and minimizing heat transfer. Within this section:

- **The diffuser:** slows down exhaust gas velocity to recover pressure, optimizing gas flow.
- **Turning vanes:** guide gases into the exhaust plenum, ensuring efficient expulsion.
- **This separation:** prevents excessive heat transfer to the generator, preserving optimal operating conditions and preventing potential damage.

4.Generator compartment

The generator compartment is where mechanical energy from the turbine converts to electrical power:

- Torque from the turbine rotates the generator's rotor, inducing electrical current in stator windings.
- A lift oil system lubricates bearings, ensuring smooth operation and reducing friction.
- Control mechanisms regulate electrical output and stabilize the power grid.
- Heat dissipation systems manage heat generated during electrical production to maintain optimal temperatures.

2.1.1.2 Steam Turbine

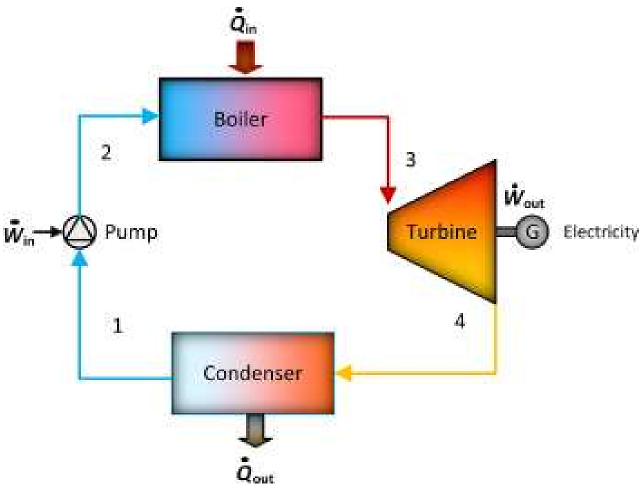

Figure 17 Rankine Cycle

1 to 2:

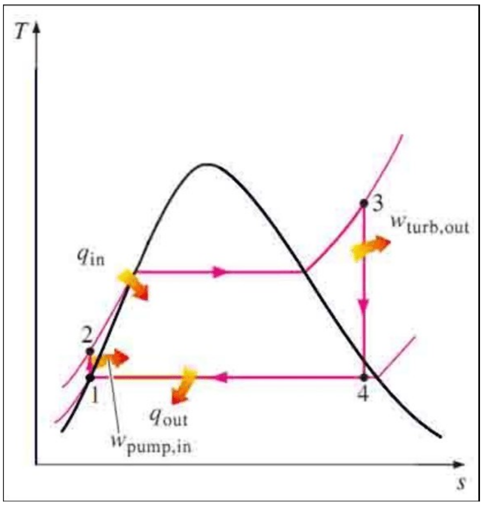

Figure 16 T-s Diagram of the Rankine Cycle

1 to 2: Isentropic compression in the feed pump

2 to 3: Constant pressure heat addition in the boiler

3 to 4: Isentropic expansion in the turbine

4 to 1: Constant pressure heat removal in the condenser

The steam turbine converts high-pressure, high-temperature steam into mechanical energy:

- Steam first converts heat energy into kinetic energy via nozzles.
- Steam then interacts with rotor-mounted blades or buckets to generate mechanical force.
- The GE SC5 steam turbine features a 20-stage design, with high-pressure (93 bar, 510°C) and low-pressure (6.7 bar, 219°C) inlets extracting steam energy.

- After exiting at 0.8 bar and 41°C, steam undergoes condensation in a water-cooled condenser using seawater, cooling it into condensate.
- Two cooling towers further lower seawater temperature, while a vacuum ejector maintains 0.8 bar condenser pressure by removing non-condensable gases.
- Powered by superheated steam from auxiliary boilers, a condensate extraction pump moves saturated water from the condenser to the LP drum through an LP heater and economizer at 25 bar pressure and 118 kg/s flow rate.

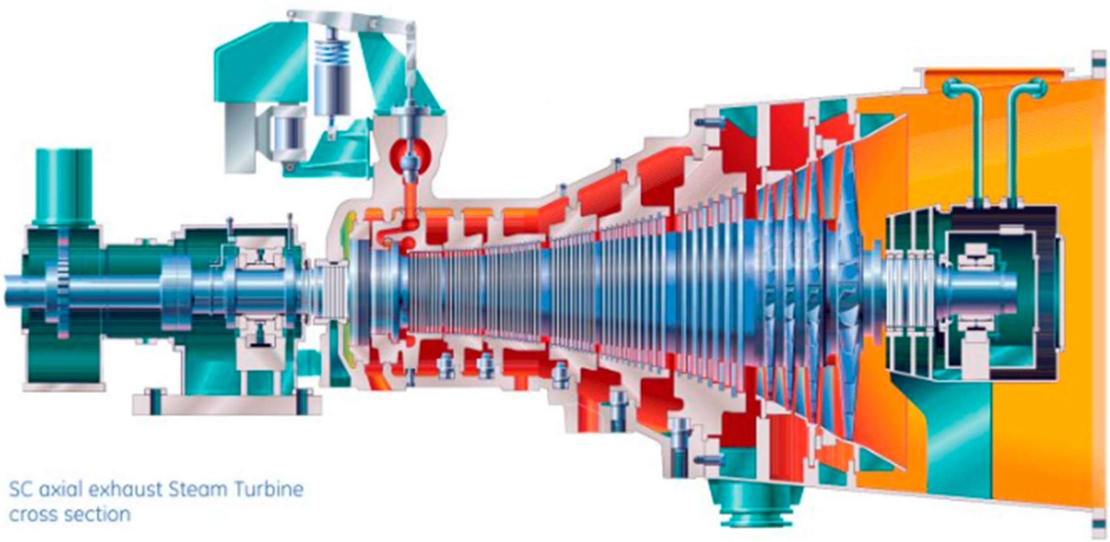

Figure 18 GE SC5 Steam Turbine

Steam Turbine Technical Data

- Brand: GE SC5
- Net Output: 110000 kW
- Stages: 20 (LP – 5, HP – 15)
- HP Steam Admission: 1st stage
- LP Steam Admission: 15th stage
- Extraction Steam: 17th stage

The steam turbine extracts mechanical energy from high-pressure steam:

- **Turbine:** High-pressure steam enters and expands through stages, rotating the shaft for power generation. The condenser reduces exhaust steam heat, enabling water condensation:

- **Condenser:** Seawater cools steam in tubes, ensuring efficient heat transfer via cooling towers. The condensate extraction pump (CEP) elevates condensate temperature and pressure:
- **CEP:** transfers essential condensate from the hot well to the LP drum, enhancing boiler efficiency. Flash boxes and tanks manage condensate drainage:
- **Closed flash boxes connect to the condenser:** while open flash tanks release gases. The air removal system boosts cycle efficiency:
- **Ejector:** Hogging removes air pre-startup and during anomalies; holding manages continuous operation gas removal.

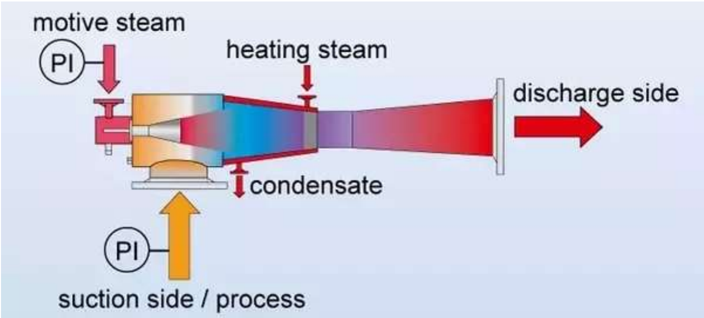

Figure 19 Super-Heated steam Ejector mechanism

- **Gland Seal System -** Protecting the turbine's bearings and preventing steam leakage is critical for smooth operation. The gland seal system supplies sealing steam to the turbine shafts, ensuring optimal performance and safety.

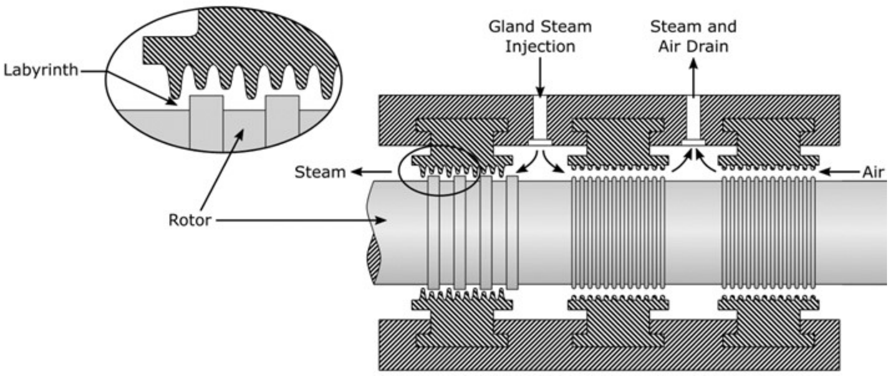

Figure 20 Gland seal system

Figure 21 Labyrinth

2.1.1.3 Heat Recovery Steam Generator

The YUGADANAVI power plant utilizes a vertical heat recovery steam generator (HRSG) to produce super-heated steam at 7 bars and 93 bars. De-mineralized water from a Water treatment plant is crucial to the HRSG's high-pressure, high-temperature operations. The HRSG includes both High-Pressure (HP) and Low-Pressure (LP) drums, along with five tube sets for effective heat exchange. Steam from the heat exchanger condenses into water in the condenser, collected and pumped by a condensate extraction pump to the LP heater. Here, it is heated before entering the LP drum, with the LP pre-heater at the top regulating this process. The HRSG's temperature ranges from 520 degrees Celsius at the bottom to 150 degrees Celsius at the top, with a pump preventing sulfuric acid formation by recirculating condensate from the LP pre-heater outlet to its inlet.

In the LP drum, incoming condensate arrives at 21 bars and 170 degrees Celsius, causing flash due to pressure drop upon entry. To prevent this, condensate must enter below its saturated temperature at 7 bars. A three-way condensate bypass valve adjusts based on LP drum saturated temperature. Once filled, condensate from the LP drum moves to the HP drum via the HP economizer, facilitated by a multi-stage centrifugal pump, ensuring efficient steam generation.

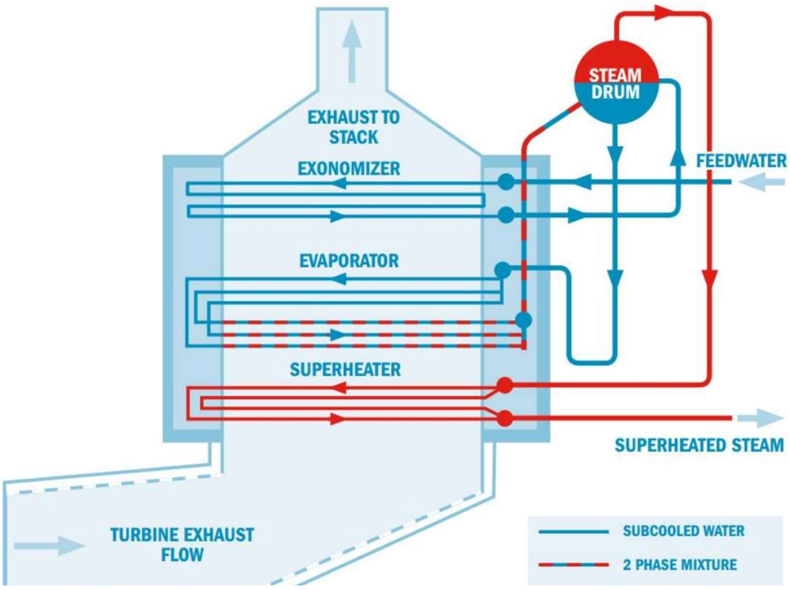

Figure 22 Vertical HRSG

The remaining condensate flows into the LP evaporator, absorbing heat from flue gas to produce steam. This steam is then circulated back to the LP drum through the condensate using natural circulation. Exiting the LP drum above the water surface, the steam passes through the LP superheater, becoming superheated before being delivered to the low-pressure stages of the steam turbine. Condensate also enters the HP drum via the HP economizer, where it undergoes steam generation in the HP evaporator and natural circulation into the HP drum. The steam then proceeds through the HP superheater, producing superheated high-pressure steam for the high-pressure stages of the steam turbine.

Condenser => LP Preheater => LP economizer => LP drum => Condensate to LP evaporator => Steam to LP drum by natural circulation => Saturated steam to LP super heater => LP SH steam to steam turbine.

The heat recovery system revolves around two main vessels: the Low Pressure (LP) drum and the High Pressure (HP) drum. These drums serve as reservoirs and mixers, blending water and steam at varying pressures and temperatures.

In the LP Drum, lukewarm water from the condenser enters, mixing with steam to stabilize at 7 bars and 167°C. It then passes through three tube bundles:

- **Economizer:** Gradually raises the water temperature in preparation for the evaporator stage.
- **Evaporator:** Intensifies the heat, causing the water to boil and transform into steam.
- **Superheater:** Elevates the steam temperature, preparing it for efficient turbine power.

The HP Drum hosts steam and water at a higher pressure of 100 bars and temperature of 311°C:

- **Economizer:** Further boosts the water temperature before entering the core process.
- **Evaporator:** Under increased pressure and heat, converts water into high-pressure steam.
- **Superheater:** Provides a final temperature increase, ensuring the steam is highly energized and ready for optimal turbine performance.

Water initially enters the LP drum, progresses through the evaporator, and a portion moves on to the HP drum where it reaches higher temperatures and pressures. The remaining steam from the LP drum continues to the turbines, driving continuous cycles of heat and pressure orchestrated by these two drum masters.

Soot Blowing system

Soot blowing is like giving a good cleaning to the outside of the tube bundles in the Heat Recovery Steam Generator (HRSG). The heavy oil used in gas turbines has sulphur, which sticks to the outside of water tubes and makes it harder for them to transfer heat. To fix this, there's a regular cleaning routine called soot blowing, usually done every 8 hours. During soot blowing, high-pressure superheated steam is sent from the high-pressure superheater tube set to the outside of the tube bundles, getting rid of the stuff that's built up.

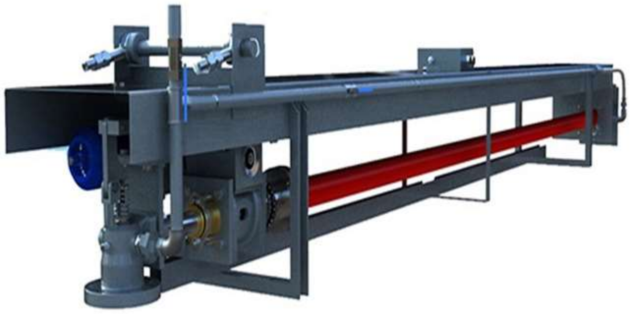

Figure 23 Soot Blower

2.1.2 Auxiliary Systems

2.1.2.1 Unloading Area

Yugadanavi Combined Cycle Power Plant (CCPP) mainly receive fuel from bowsers. Bowsers supply fuel to seven bays, with six dedicated to Heavy Fuel Oil (HFO) and one for Diesel Oil (DO). Each bay has two unloading pumps with a capacity of 16.5 cubic meters per hour. Daily requirement of bousers is 47, when the plant runs in full load.

Figure 24 Unloading Area

2.1.2.2 Tank Yard

The unloaded HFO is transferred to two 10,000 cubic meter tanks. To prepare the HFO for the gas turbine, it undergoes treatment by the Fuel Oil Treatment Plant (FOTP) to remove contaminants like Sodium, Potassium, and sludge. Following treatment, the oil is transferred to three 1,500 cubic meters Treated HFO tanks. Additionally, there are two DO tanks with a capacity of 2,500 cubic meters each. The FOTP has an intake capacity of $75m^3$ /h, which is divided into four lines of $25m^3$ /h, each three and one line is at standby position. These lines guide the fuel through two mixers and two separators in each, ensuring a thorough treatment process.

2.1.2.3 Fuel Oil Treatment Plant

The Fuel Oil Treatment Plant (FOTP) is crucial for gas turbine operation, focusing on removing water-soluble contaminants from untreated Heavy Fuel Oil (HFO). It also treats wastewater from the process and separates diesel oil using separators. The plant buys HFO with sodium + potassium concentrations below 30ppm due to separator capabilities.

FOTP employs four unit sets for separation, each comprising two mixers and centrifugal separators. It ensures fuel quality assurance, prevents equipment damage, optimizes combustion, enhances system efficiency, extends equipment lifespan, and maintains reliability, availability, and environmental compliance.

The untreated HFO is initially preheated using hot water from an auxiliary boiler, raising its temperature to 55-60°C. Two screw-type pumps then transfer the preheated oil (75m^3/h capacity), with one pump on standby for filter blockages. Two filters remove solid particles, followed by heat exchangers that raise HFO temperature to 70°C and then 90°C using treated HFO and hot water (120°C coil temperature).

A Demulsifier (GT 217) aids HFO and water mixing before entering mixers with internal coils. Mixed HFO and water then undergo separation in Alfa Laval separators, producing treated HFO, wastewater, and sludge after passing through two separator units.

Maximum contamination limits

	Na + K	**Pb**	**V**	**Ca**	**Al/Si/Mg**
HFO	<1 ppm	<1 ppm	<100 ppm	<10 ppm	Report if above <5 ppm
DO	<1 ppm	<1 ppm	0.5	<2 ppm	Report if above <5 ppm

Table 1 Maximum contamination limits

After separation, treated Heavy Fuel Oil (HFO) goes to untreated tanks, while sludge and wash water go to the sludge basin. Dedicated pumps transport sludge and oily water to the waste treatment unit. sludge is piped to the reclaim tank, and then goes to the dirty water tank. Oil from the reclaim tank is pumped back to untreated HFO pipelines for reprocessing. Water in the dirty water tank undergoes skimming, with oil sent to reaction tank 1. Water from reaction tank 1 moves to reaction tank 2 via a U tube, then to the pre-cleaned tank and water separators. Separated water flows to the Effluent Treatment Plant (ETP), while separated oil goes to the basket centrifuge sludge bin. Sludge from the separator goes to the separator sludge bin, combined with oil for centrifuge processing.

2.1.2.4 Sea Water Intake Plant

Sea water for the power plant is purified through a series of treatments to remove particulate waste. Initial treatment occurs at the sea water intake plant, where water is extracted and pumped to the plant. The process begins with a sand trap to remove large particles, followed by a bar screen with a grab cleaner to capture debris and prevent clogging. Next, water passes through a revolving chain screen and multidisc screen to eliminate finer impurities. Raw water pumps then draw the pretreated sea water into the power plant. The purification operates in two paths controlled by stop log gates, allowing cleaning in one path while purifying in the other, managed by a crane-operated system.

Figure 25 Bar screen

Figure 27 Revolving chain screen

Figure 26 Multidisc screen

2.1.2.5 Water Treatment Plant

The Water Treatment Plant (WTP) produces Demineralised (DM) water and service water from seawater. DM water is used in HRSGs, CCW system, and aux. boiler, while service water serves other plant needs. Sodium Hypochlorite (NaOCl) is added to seawater at the WTP to prevent marine growth before it enters the Sea Water Treatment Plant (SWTP). NaOCl is generated via electro-chlorination. Coagulants like Aluminium Sulphate and Poly Aluminium Chloride are added in the flash mixer to neutralize suspended solids, followed by flocculants in the flocculation tank to bind particles. Tube settlers trap solids, and clarified water undergoes dual media filtration. Ultrafiltration units remove remaining suspended solids. Sea Water Reverse Osmosis eliminates dissolved solids by pressurizing water through a semi-permeable membrane. A second Reverse Osmosis unit, Brackish Water RO, further purifies water, significantly reducing conductivity.

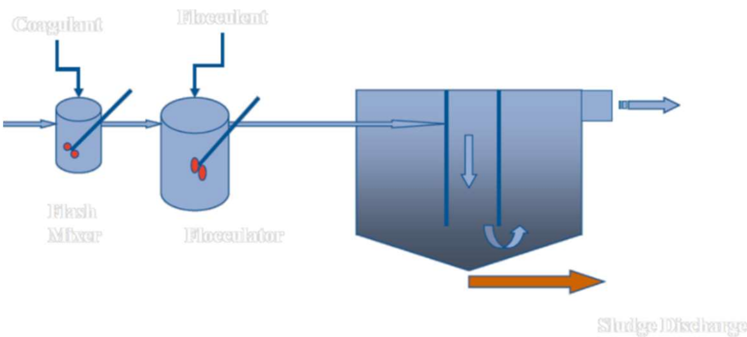

Figure 28 Pretreatment process

Purified water from the BWRO unit goes to mixed bed ion exchangers, featuring anion and cation beds. These beds require periodic regeneration. The demineralized water is stored in a dedicated tank.

Figure 29 Dual Media Filters

Figure 31 Ultra Filtration

Figure 30 Sea Water Reverse Osmosis

Figure 33 Brackish Water Reverse Osmisis

Figure 32 Mix Bed Exchanger

2.1.2.6 Closed Cooling Water Plant

CCW System is a closed cycle cooling system which uses DM water to cool several accessories of the plant. In this system, heated water passes through the heat exchanger and then cooled water is pumped by the cooling water pumps. This setup is referred to as a closedcircuit system, designed to be operational during plant start-up, normal operation, and shutdown. The system is filled with DM water to which corrosive inhibitors are added. Seawater is used in the heat exchanger to cool the heated DM water.

2.2 <u>Training Experience</u>

2.2.1 Safety Practices

At the Yugadanavi Power Plant, there is a high concern for the safety of their employees, visitors and persons involved in their third-party operations. Mandatory Personal Protective Equipment (PPE),

- Safety shoes
- Safety helmets
- Hand protection, ear protection, and eye protection equipment must be used when necessary

Figure 34 PPE

Safety Procedures being followed,

- Overalls or trousers up to ankle length must be worn.
- Training programs and fire drills are done at least once a month to be ready for any accidents or hazards.
- Smoking inside the power plant is strictly prohibited.
- All accidents and illnesses must be immediately reported to the safety officer.
- A Work Permit should be issued from the control room before the commencement of any work inside the plant.
- Safety goggles must be worn for gridding, lathwork handling and chemical dosing work
- Relevant operating instructions must be followed when operating a machine
- All the safety signs should be strictly adhered to
- Informing injuries, and reporting near misses are made compulsory.

2.2.2 Mechanical Maintenances Involved

2.2.2.1 Maintenance in Aux. Boiler Diffuser

The Auxiliary Boiler is crucial for generating steam used to heat Demineralized (DM) water for various purposes in the Gas Turbine (GT), Steam Turbine (ST), Tank Yard, and Fuel Oil Treatment Plant (FOTP). It operates as a three-pass wetback smoke tube boiler with a capacity of ten tons, producing 5.39 × 10^6 kcal/hr. of heat at a pressure of 10.54 kg/cm^2, using Heavy Fuel Oil (HFO) as fuel.

Maintenance was performed on the boiler's burner gun, which includes a fuel nozzle, ignition system, flame scanner, and diffuser. The burner operates as an industrial pressure jet with ignition by high-voltage spark.

During maintenance, it was discovered that the diffuser plate, responsible for even air distribution in the combustion chamber, was bent and had accumulated carbon deposits, causing reduced steam production. The maintenance procedure involved stopping the boiler, allowing it to cool, removing and cleaning the combustion head, specifically the diffuser plate, correcting any bends using appropriate tools, potentially replacing the fuel nozzle, and reinstalling all components securely.

2.2.2.2 Soot Blower Gland Replacement

Due to the flue gas, carbon is deposited on the outer surface of the HRSG tubes, which can reduce the efficiency of the steam generation process. So, soot blowers are used to remove this deposited carbon.

There are three main types of soot blowers,

- Furnace wall soot blowers
- Stationary soot blowers
- Long retractable soot blowers

From them, the long retractable soot blower method is used in this power plant.

Maintenance

- First checked any pressure or steam within the blower was relieved.
- Then nuts and bolts were removed. The hoses and pipes that were connected to the gland were removed.
- The old gland was carefully removed and inspected the surrounding area for any sign of damage.
- Then new gland was replaced into the housing and gasket was placed. Removed pipes, nuts and bolts were also placed.
- Then performed an inspection to ensure that all components were seated.

Figure 35 Soot Blowers

Figure 36 Soot Blowers

2.2.2.3 HRSG Expansion joint leak Insulation

In HRSG assembly, expansion joints (bellows) are situated at the inlet duct. They are typically made from metal or fabric materials and feature a convoluted or corrugated design to provide flexibility while maintaining pressure integrity. There are some functions of these expansion joints.

- Allow for the expansion and contraction of the HRSG inlet ducting due to temperature variations. Also allow movements in multiple directions.
- Help to absorb and dampen vibrations generated by the gas turbine operation.
- Prevent exhaust gas escaping from HRSG inlet duct.

In HRSG 2, an exhaust gas leak occurred in the expansion joint. The maintenance procedure involved the following steps,

- Initially, the leaking areas in the joint were located and identified.
- Subsequently, power generation from GT02 was halted.
- The previously identified areas were covered using a new bellow, and nuts and bolts were used to secure it in place.
- Further insulation was applied by sealing the ends of the joint with glass wool.
- Finally, an inspection was conducted to ensure there were no remaining leaks.

Figure 37 Some pictures of the insulation process

2.2.2.4 Bearing replacement in BT fan Motors

The BT fan motor in GT01 experienced a fault, impacting its crucial role in heat removal from the turbine compartment. The issue involved a broken bearing and damaged motor winding. To address this, the fan system was carefully removed along with its cover using a crane. A specialized 6313 ZZ C3 bearing designed specifically for the BT fan motor was selected for replacement, ensuring compatibility and optimal performance.

Effective repair of the fan motor is essential for maintaining the gas turbine's cooling operations, which are vital for regulating temperatures within components like combustors, turbine housing, and blades. This cooling process is pivotal for enhancing overall turbine efficiency and ensuring reliable electricity generation.

Figure 39 BT fan removing process

Figure 38 Bearing of the BT fan motor

2.2.3 Experiences with Measuring devices and Operational tools

2.2.3.1 Lubricating bearings in the motors of the gas turbine plant using the Lube Expert device.

In motors, bearings prevent damage by reducing friction with lubrication, which creates a protective layer between moving parts. This minimizes heat generated by friction, crucial for large motors to prevent overheating and reduce vibration and noise during operation.

In gas turbines, lubrication is applied to turning gear, AC lube oil pump, TK, VG, and EF fan motors. The Lube Expert SDT270 device checks lubrication levels by analyzing machine sounds, especially high-frequency signals, to assess machine condition. Initial dBµV readings provide RMS, maxRMS, PEAK, and Crest Factor measurements, guiding lubrication needs.

Three methods are used to set lubrication baselines:

1. Comparison: Similar bearings are tested simultaneously to establish dB levels.
2. Lubrication Monitoring: Grease is applied until the dB level stabilizes, indicating optimal lubrication.
3. Historical Data: Initial dB levels are compared with readings after 30 days to set baseline levels.

From these methods, the second method was used. First got the value and applied grease and then checked about that value. If it reduced, the lubrication was done when it stops decreasing.

Figure 40 Lube Expert SDT270 Device

2.2.3.2 Vibration monitoring in cooling tower motors.

This program was conducted as a condition-based monitoring program. The FALCON 01dB vibration monitoring device was used to check the vibrations. The FALCON 01dB is an appropriate instrument designed for this purpose, equipped with a range of features to facilitate comprehensive vibration analysis. The following are some of its features,

- The device is equipped with high-precision accelerometers that can detect even the slightest vibrations.
- FALCON 01dB supports multiple channels, enabling simultaneous monitoring of different parts of a machine.
- The device includes Fast Fourier Transform (FFT), to analyse the frequency components of vibrations. This helps in identifying specific issues like imbalance, misalignment, bearing faults, and gear defects.
- It is suitable for on-site monitoring and can be easily transported between different locations within a facility.

Procedure: -

- The triaxial accelerometer and the FALCON 01dB device were chosen for the application.
- The critical points were determined on the motor where vibration measurements will be taken.
- The accelerometer was attached to the predefined monitoring points on the machine.
- Configuring parameters such as sampling rate, measurement duration, and frequency range were Set up the FALCON 01dB.
- Vibration measurements were taken and stored in the device.

Figure 41 Falcon 01dB Device

2.2.3.3 Valve Condition monitoring Using Ultrasonic

Valves are critical for gas, oil, and water transportation, regulating flow and preventing energy loss or product quality issues when functioning improperly. Ultrasonic inspection using the SDT 340 device identifies valve conditions like leaks or blockages through a condition-based monitoring program.

The method involves:

1. Identifying system details (normally open or closed).
2. Assessing valve condition by touching points upstream (A and B) and adjusting sensitivity until the display reads 50%.
3. Comparing sound levels downstream (C and D):

    - A louder sound at C than B suggests fluid passage.
    - Low sound at downstream C indicates a closed valve.
    - Louder sound at D compared to C indicates a leak further downstream.

This method swiftly identifies valve functionality and potential issues using ultrasonic technology for effective maintenance and operation management.

Figure 43 Reading for a closed valve

Figure 42 Reading for an open valve

2.2.3.4 Gate valve seat grinding using Ventil Orbit 10 Valve grinding machine.

Over time, valves and valve seats corrode, hindering proper sealing between valve seats. Consequently, valve leakage occurs. Therefore, if they are worn unevenly or have any imperfections, grinding may be necessary to restore a smooth surface for proper sealing.

For this purpose, the Ventil Orbit 10 valve grinding and lapping machine was utilized. This machine can be used both horizontally and vertically and can accommodate valve diameters ranging from 1.5 inches to 10 inches. The machine operates pneumatically, requiring a minimum air pressure of 7 bars.

Procedure:

- In this maintenance task, a 6-inch gate valve was utilized, necessitating the use of the vertical arm of the machine.
- Initially, the valve was cleaned and placed on the table. The vertical arm of the machine was then secured, and 100µm roughness stickers were applied.
- Next, the machine was fixed onto the valve, and the air supply was connected.
- Starting from a low speed, the procedure was initiated, and after a few minutes, the machine was stopped to inspect the valve seat.
- This procedure was repeated three times using 100µm, 60µm, and 30µm roughness stickers, respectively.
- Following this, the valve seat was checked for improved sealing, and then the machine was disassembled.

Figure 44 Ventil Orbit 10 Device

2.2.3.5 Measuring The thickness of the drums, tube bundles and headers in HRSG 02

The HRSG consists of about 1000 tube bundles and headers. The primary objective of this program was to assess whether the drums, tubes, and headers had experienced any wear over time. Based on a previously created graph, thickness measurements were taken at random using the Olympus 38DL Plus ultrasonic device, equipped with a single-element transducer designed for highly precise measurements of thin or multilayer materials.

Procedure: -

- The single element transducer was connected according to requirement.
- The device was calibrated using a calibrator plate.
- According to the previously made graph, the areas were marked before taking the measurements.
- Water or grease was used to the face of the transducer to remove air gap between the transducer and the surface.
- Measurements were taken.

Following are the average thickness values,

- HP drum hemispherical part = 60mm
- HP drum cylindrical part = 90mm
- LP drum hemispherical part = 25mm
- LP drum cylindrical part = 25mm
- LP side tubes = 3-4 mm
- LP side headers = 12 – 13.5mm
- HP side tubes = 3-5 mm
- HP side headers = 20,25 and 30mm

Figure 45 Olympus 38DL Plus

2.2.3.6 Shaft alignment using EASYLASER device.

Proper shaft alignment reduces the risk of mechanical failures, increases efficiency, and lowers operational costs. Therefore, a shaft alignment program was conducted for the cooling tower fans, specifically for the shaft between the motors and the gearbox. The Easy Laser XT660 device was used for this purpose.

The Easy Laser XT660 is a laser alignment system designed for industrial applications. It is utilized primarily for the precision alignment of machinery, ensuring optimal performance and reducing wear and tear. This device can measure shaft alignment, straightness, flatness, and more, making it versatile for various alignment tasks. This device is equipped with wireless technology and allows for easy data transfer and remote monitoring.

Procedure: -

- First, we ensured that all machines were stopped according to safety protocols.

- Laser transmitters and detectors were mounted on the shaft using chain brackets.
- The horizontal shaft alignment program was selected from the device's interface, and the dimensions, such as the distance between the measurement points, were entered.
- Initial measurements were taken while rotating the shaft manually. The data were captured at specific points, typically at 9, 12, and 3 o'clock.
- The current misalignment was calculated automatically by the device after collecting data from each point.
- The position of the motor was adjusted based on the alignment data. Shims under the motor feet were used to change the position of the motor in vertical direction. This was done by adding and removing shims with various thicknesses. The horizontal plane alignment was done by moving the motor in horizontal direction. Eight bolts were used for that.
- After the adjustment, the alignment was re measured and improvements were checked.
- This measurement and adjustment process was done until the shafts were aligned within acceptable tolerances.

Figure 46 Easy Laser XT660 Device

2.2.3.7 Dye Penetrant Testing

Dye penetrant inspection is a method where inspectors use a dye or liquid to detect surface defects on non-porous materials like ceramics, plastics, and metals. The process relies on capillary action, where the dye penetrates surface cracks and discontinuities. The inspection involves three main steps: cleaning the surface, applying red dye penetrant, and then applying a developer.

Inspectors use this method to examine welds, castings, forgings, plates, bars, and pipes for defects such as leaks, joint flaws, fractures, porosity, and various types of cracks (fatigue cracks, hairline cracks, grinding and quenching cracks).

Benefits of dye penetrant inspection include ease of use on complex surfaces, cost-effectiveness (no need for expensive equipment), rapid inspection of large areas, and clear visual indication of defects on the material's surface, including the defect's dimensions. This method is widely applicable across different industries for surface defect detection.

This dye penetration test was conducted to the Steam turbine rotor, couplings to identify the cracks in outer surface.

Procedure:

- First, the surface plan to test was cleaned using cleaner. So that the surface is open and any defects it contains will be exposed, instead of remaining hidden underneath dirt or other foreign elements.

- The red dye penetrant was sprayed on the required surfaces in the rotor and the coupling and gave around 15 minutes to it to dry.

- After cleaning off extra penetrant, cleaner was again applied to the surface and rubbed it dry with a fresh clean, dry rag.

- After cleaning and removing the dye penetrant, the white developer applied it to the surface. The developer will draw the penetrant from the flaws or cracks on the surface of the material and make them visible.

- At this point, cracks and other types of defects will be visible. There were not any crack or defect in turbine rotor and coupling.

Figure 48 The developer, the red penetrant and the cleaner

Figure 47 Dye penetration testing

2.2.4 Projects Involved

2.2.4.1 Fireline Endcaps casting using sand casting method

Casting Process

Manufacturing Fireline endcaps was begun to meet the requirements of our power plant, utilizing aluminium and employing the sand-casting method.

Two sand casting methods were experimented. In the first approach, a plate with a hole was used, matching the cross-section through the centreline of the pattern. In the second method, casting using two halves of the pattern was used. Wooden patterns were employed in both methods, crafted to the required dimensions using a lathe machine.

The procedures for both methods were identical. Initially, water to the sand was applied and mixed it, then proceeded to create the mould. Sprues were utilized for pouring aluminium, while risers ensured the mould was adequately filled. Gate systems, runners, and vent holes were added as necessary. Additionally, a core was prepared separately and incorporated into the mould.

Discarded aluminium found in the plant was melted using a furnace. Subsequently, the molten aluminium was poured into the mould until it overflowed through the risers. After a few minutes, the mould was removed, and the component was washed. Machining was employed for achieving a superior surface finish.

Figure 50 Casted Endcaps

Figure 49 Casted Endcaps

Pattern Designing

I had another task to design a pattern for those end caps using SolidWorks for 3D printing. Initially, I gathered all the necessary dimensions and drew a plan view and a front view of the pattern. Subsequently, a plate with three series of holes and three series of patterns was designed, all tailored to the required size. This plate is employed for manufacturing end caps using the previously mentioned first method. Additionally, four more patterns were created, each with two halves, specifically designed for the second method.

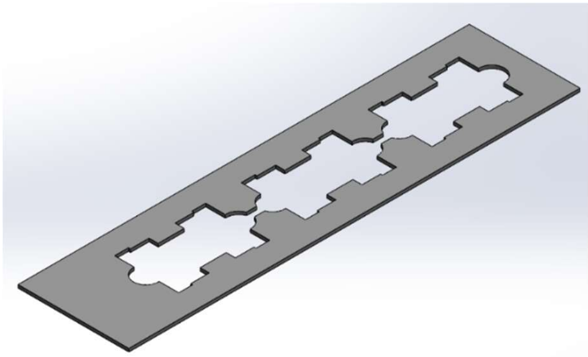

Figure 51 A plate design for method one

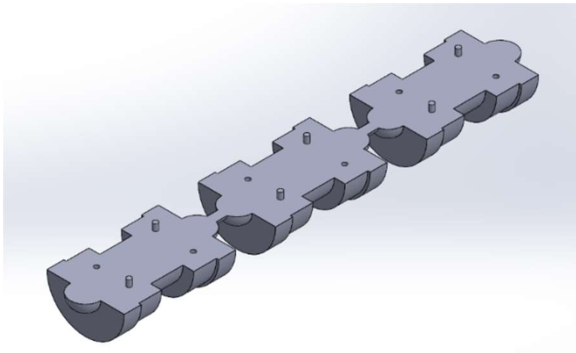

Figure 52 One half for method two

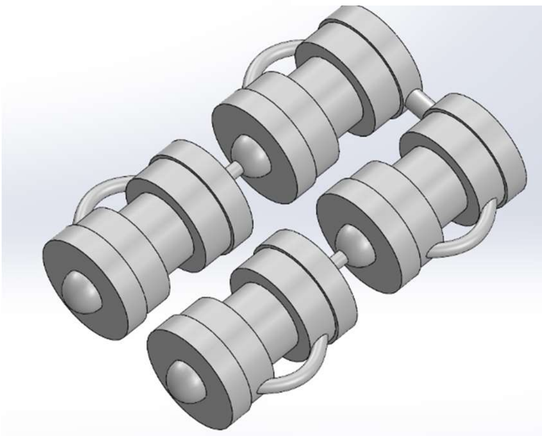

Figure 53 Four patterns for method two

2.2.4.2 Designing and fabricating of A structural support for the steam turbine upper casing

This structure was designed to place the upper casing of the steam turbine.

- The design idea was discussed with the engineers and the required measurements were taken using measuring tape and level meter.
- Then the 2D design was sketched in a paper to get an idea.

- The 3D design was designed using SolidWorks software according to the previous sketches.
- For the structure, 100mm size I beams and C channels were used due to their high strength and there were many scrap parts in Yugadanavi Power Plant.
- C channels were used for the cross beams because there were no horizontal forces. They were used as supporters for the columns.
- A simulation was done for the whole structure using SolidWorks software with the safety factor of 2. The used force was 60000N according to the weight of the upper casing.
- The maximum stress was 3.398x105 and that is less than the yield strength (3.51x108) and the maximum displacement was 0.0008811mm on the top plates. Therefore, the structure was stable according to the design.
- Finally, fabrication was done with the help of fabricators in this plant.
- Mig welding was used to the fabrication due to its higher welding strength.

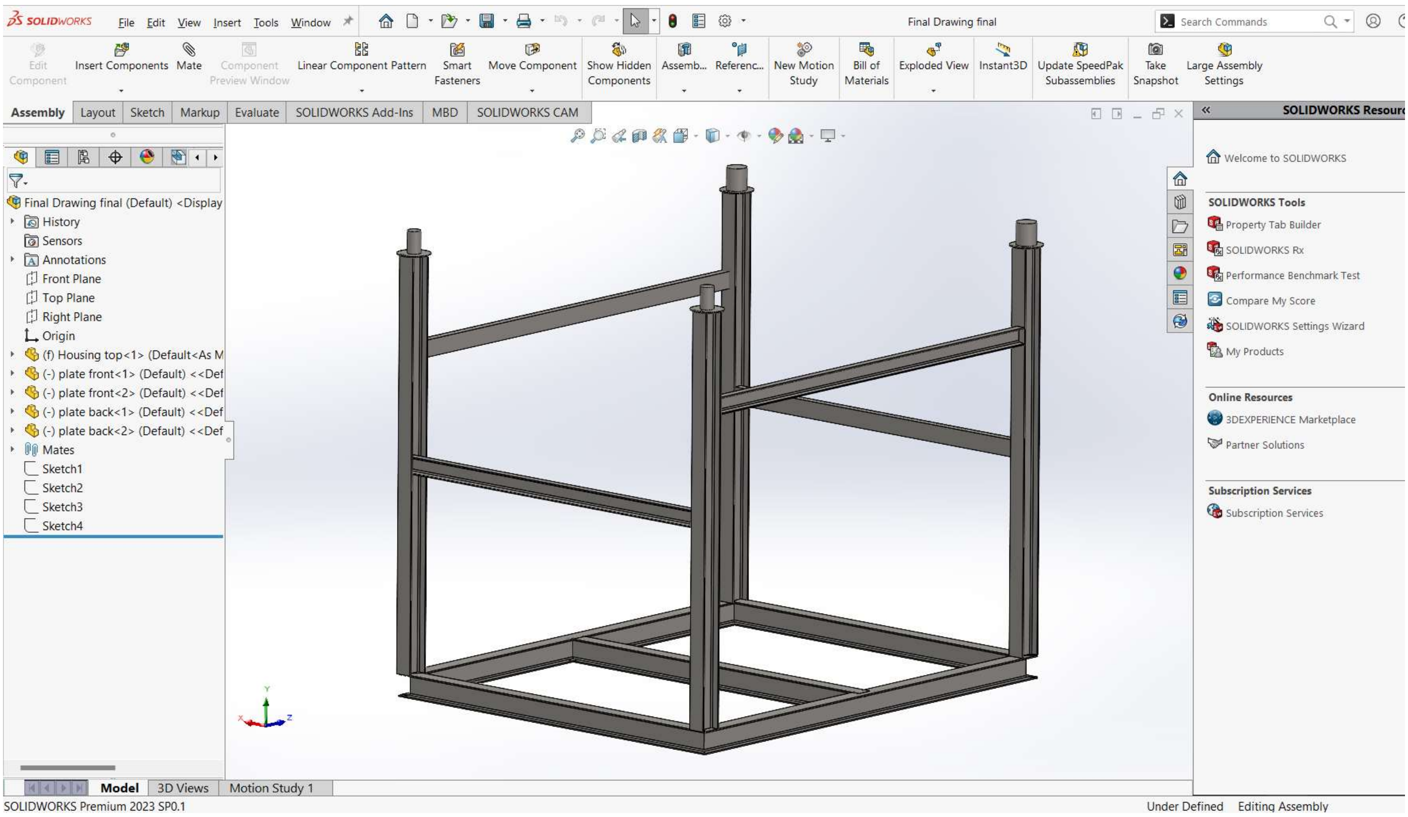

Figure 54 SolidWorks 3D design

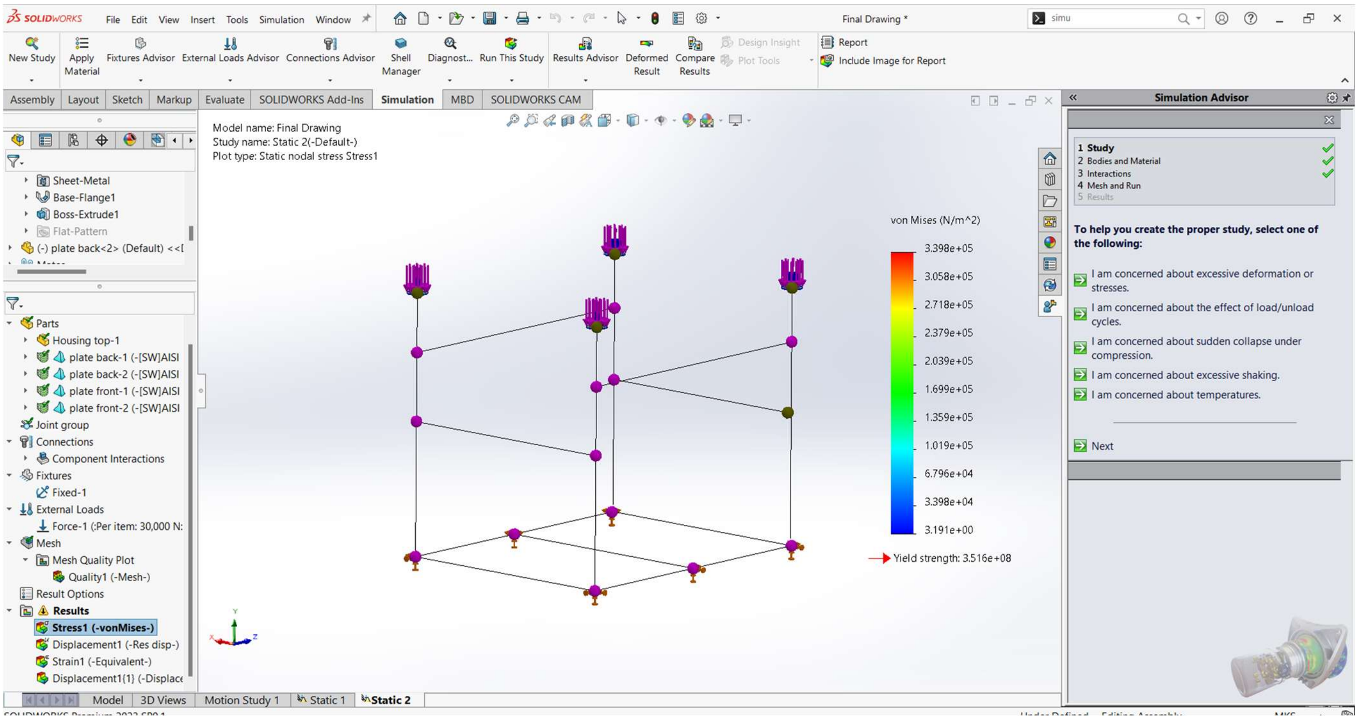

Figure 56 Stress analysis for the structure withstand for 12Tons

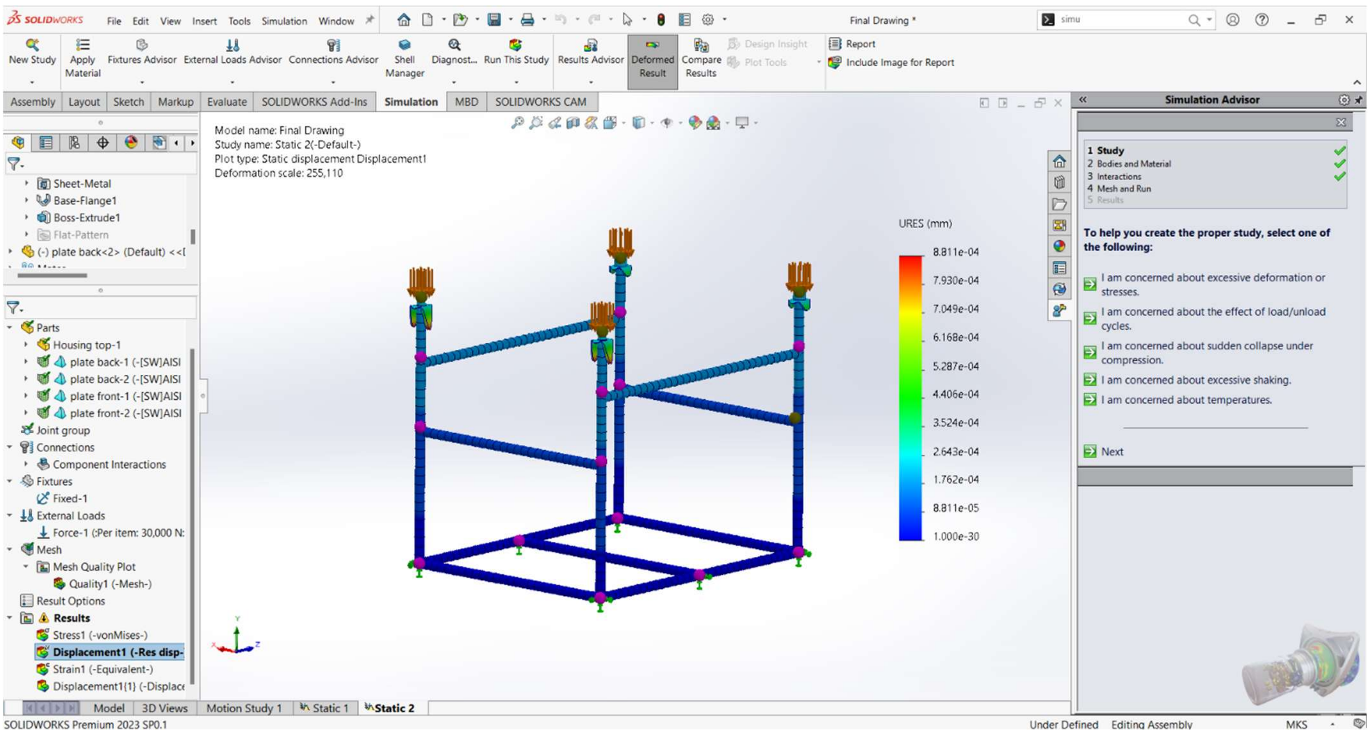

Figure 55 Displacement analysis for the structure

Figure 57 Fabricated structure

Figure 58 The structure with the ST upper casing

2.2.4.3 Identification of failure types of Bearings using visual Inspections

Bearings are used to constrains relative motion into desired motion. These machine elements are made for a specific lifetime. But some of these bearings fail before reaching its lifetime. This situation is called as bearing failure. There are many failure types and reasons for that failures. Preventing from that increases lifetime of the bearings.

First, I had studied about the bearing failure types and reasons for them. Then a report about bearing failure type was created. According to that report, I had studied about five nearly failed roller bearings in this power plant.

1. Cooling tower 11 fan motor bearing.
2. Condensate recirculation pump 2 motor Non drive end bearing.
3. Condensate recirculation pump 2 motor drive end bearing.
4. GT 2 water injection pump motor Non drive end bearing.
5. GT 2 water injection pump motor drive end bearing.

I had collected failure types using visual inspections from every bearing. Then the reasons for the failure were identified and created reports for every bearing including countermeasures for them. Here are the some of them. Examples for the inspections are available in **Annex 01**.

Bearing	**CT fan**	**CRP NDE**	**CRP DE**	**Water injection pump NDE**	**Water injection pump DE**
Failure Types	Wear Movements of the running trace Fretting	Fretting Wear	Fretting Scoring Wear Pitting Corrosion	Cage damage. Wear Scratching	Wear
Reasons for the failures	Entry of foreign matter. Improper or insufficient lubrication. Sliding due to irregular motion of rolling elements. Inner ring or outer ring rotation Axial load in one direction. Poor lubrication Vibration with a small amplitude Insufficient interference	Poor lubrication Vibration with a small amplitude Improper or insufficient lubrication.	Sliding due to irregular motion of rolling elements. Excessive load, excessive preload. Inclination of inner and outer rings. Poor lubrication. Particles are caught in the surface. Exposure to moisture in the atmosphere.	Extraordinary vibration, impact, or moment. Improper or insufficient lubricant. Improper or insufficient lubrication. Insufficient lubricant at initial operation. Careless handling.	Entry of foreign matter. Improper or insufficient lubrication.

Counter measures	Use a proper lubricant. Apply a preload. Check the interference fit. Apply a film of lubricant to the fitting surface. Proper Alignment with the shaft and mating components. Improve the sealing mechanism.	Improve the lubricant and the lubrication method. Prevent misalignment. Improve the sealing mechanism.	Check the size of the load. Adjust the preload. Filter the lubrication oil thoroughly. Use a proper lubricant. Improve the storage methods.	Reexamine the selection of lubricant or lubrication method. Minimize the mounting deviation. Improve mounting. Clean the housing. Improve the mounting procedure.	Improve the sealing mechanism. Clean the housing. Check the lubricant and lubrication method.

Table 2 Bearings failure types, reasons and countermeasures

2.3 Hot Gas Path Inspection in Gas Turbine 02

A hot gas path inspection is a critical maintenance procedure performed on industrial gas turbines to assess and maintain their performance and reliability. This inspection focuses on components exposed to high-temperature exhaust gases, such as turbine blades, vanes, combustion chambers, and associated seals. The primary objectives are to detect and address wear, damage, or degradation that could compromise turbine efficiency, output, or safety.

During this hot gas path inspection, the turbine's upper casing was removed by technicians to visually inspect components for signs of erosion, corrosion, cracking, or coating degradation. First, a dial gauge was placed under the shell before it was lifted and the change in its position was observed A 60T crane, chain blocks, Hydraulic lifting jacks, were used to lift the turbine shell. The shell was lifted to 600mm using hydraulic jacks and then the shell was lifted with the help of guide Pins and 60Tcrane.

The upper and the lower casing of the turbine were inspected and repaired. The nuts and the bolts in the turbine casing were cleaned. The following were replaced.

- The turbine nozzle.
- Turbine buckets.
- Shrouds.
- Nozzle Pins.
- Shroud pins.
- Bores cope pins

Figure 59 Turbine rotor with buckets

2.4 Combustion Inspection in Gas Turbine 02

This inspection focuses on assessing and maintaining the combustion system, including the combustion chambers, fuel nozzles, igniters, and associated components. The primary goal of a combustion inspection is to evaluate the condition of these critical components to ensure efficient and stable combustion. This involves inspecting for signs of wear, erosion, carbon deposits, and damage that can affect combustion efficiency and emissions. Issues such as hot spots, fuel distribution problems, and burner performance are also evaluated during this inspection. The following components were inspected, repaired, or replaced, as necessary.

- Fuel nozzle.
- Combustion covers.
- Flow sleeves.
- Liners.
- Purge valves.
- Check valves.
- Transition pieces.
- Bull hones.
- Cross fire tubes.

In this inspection, first all the components that were attached to the combustion chamber were removed. Then inspections were done for the components mentioned previously. Then cleaning processes and replacements were done as the requirements.

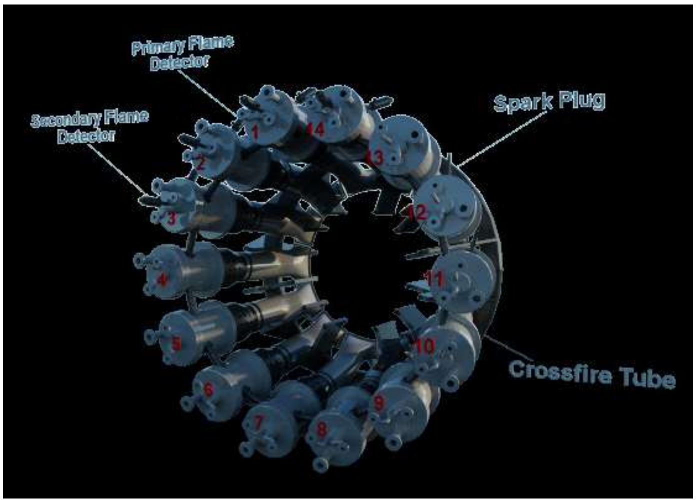

Figure 60 Combustion chambers of a gas turbine

2.5 Main Inspection in Steam Turbine

2.5.1 About MI

A major inspection of a steam turbine is a comprehensive and detailed maintenance procedure carried out to ensure the turbine's reliability, efficiency, and longevity. The primary goal is to thoroughly examine the turbine's components for signs of wear, fatigue, corrosion, and other forms of degradation that could affect performance and safety. By cleaning, repairing, or replacing worn parts, the turbine's efficiency can be restored or even improved, leading to better overall performance and reduced operational costs.

Key Components Inspected

- **Rotors and Blades**: Inspecting for cracks, erosion, and deposits that could impair functionality.
- **Casings and Diaphragms**: Checking for structural integrity and wear.
- **Bearings and Seals**: Ensuring they are in good condition to prevent leaks and reduce friction.
- **Steam Path**: Cleaning and inspecting to ensure efficient steam flow.
- **Control Systems**: Verifying the accuracy and responsiveness of control mechanisms.
- **Cleaning of the condenser and flash boxes**.

- **Gland sealing system, Condenser ejector system, Bypass valve system and Lube oil system**: Inspecting for cracks, erosion, and deposits that could impair functionality.
- **Condensate Extraction Pump:** Check the bearings.
- **Steam Control Valves:** Check for the leaks in the seatings.
- **Turbine insulations:** Replaced

Process

- The turbine was safely shut down and allowed to cool to prevent thermal stress during disassembly.
- Major components were carefully disassembled to allow for detailed inspection.
- Components were cleaned to remove deposits and allowed for better inspection accuracy.
- Techniques such as ultrasonic testing and dye penetrant inspections were used to detect internal and surface defects in rotors, bearings…etc.
- Precision measurements were taken to ensure components are within operational tolerances. Bolts and nuts were numbered before removing. Thickness values were also taken between rotor buckets and casing, gland sealing and the rotor, labyrinth seals and the rotor. Dial gauges and depth gauges were mainly used to take the measurements.
- Damaged or worn parts were repaired or replaced as necessary. Sand blasting was done for the rotor buckets and diaphragms.
- The turbine was carefully reassembled, with attention to alignment and balancing. Then rotor was rotated and checked the radial runouts using dial gauges before fixing the upper casing.
- The turbine was tested and calibrated to ensure all systems are functioning correctly.

Figure 61 Steam Turbine Rotor

The HP side casing was connected by a special type of stud. To loosen these, bolt heaters were used. The heaters were inserted into the studs and heated to a temperature of 300°C. This decreased the tightness between the studs and the casing. Then the nuts and studs were removed using a spanner wrench.

Figure 63 Bolt heater with a spanner wrench

Figure 62 Bolt heater

The condenser tubes were cleaned using tube cleaning bullets and high-pressure water gun. To successfully perform the process, cleaning bullets were first fitted into the tubes using a water gun to propel the bullets through the tubes. This made it possible for the tubes to undergo thorough cleaning to get rid of any form of deposit as well as debris that might have set in.

Figure 64 Condenser cleaning bullets

Figure 65 High pressure water gun

2.5.2 Work Involvement in MI

2.5.2.1 Bolt Removing of the coupling between generator shaft and turbine shaft using Riverhawk hydraulic Interlock Tensioner.

Coupling is essentially a mechanical device that transmits rotational motion between shafts. Hydraulic coupling bolts are commonly used in rigid coupling systems. When using a regular bolt to secure a flange, a gap between the bolt and the flange hole can cause the bolt to loosen over time. To prevent this, hydraulic coupling bolts have sleeves that fill the gap between the bolt and the flange hole. The inner surface of the sleeve and the contact surface of the bolt are designed with a specific slope to ensure a secure fit.

Method

- Bolt tensioning cylinder was threaded on by a hand ratchet.
- The high-pressure hose line had been coupled to the hydraulic pump and hydraulic pressure was applied to the tensioner.
- The nut was loosened using a spring-loaded gear drive system with a hand torque wrench.

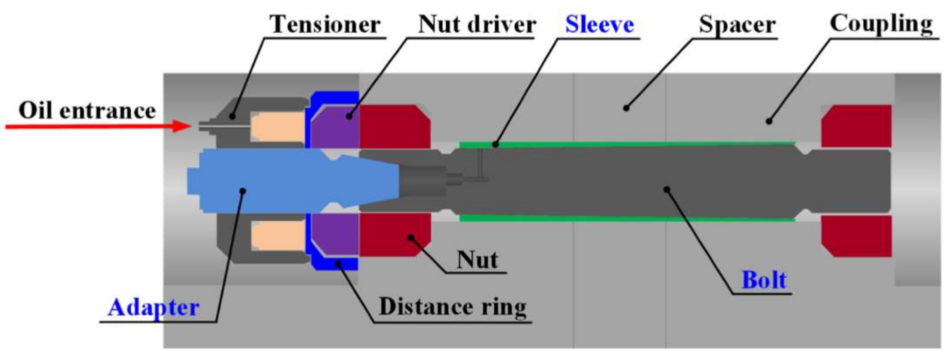

Figure 66 Nut losing method

- The pressure was then released from the bolt tensioning cylinder and the tensioner automatically returned to its starting position by means of an internal spring return system.
- Then adapter was removed, and high-pressure hose line with an oil injector were used to pump oil into between sleeve and the bolt.
- The high-pressure oil (16000psi) was loosened the bolt, and it was removed.
- Then the bolt tensioning cylinder was threaded off the stud.

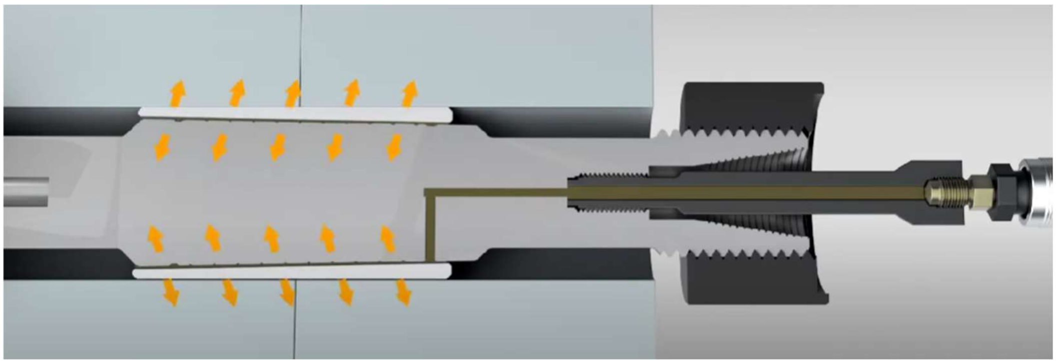

Figure 67 Stud removing method

2.5.2.2 Surface profile/roughness visual & quantitative analysis in Steam Turbine Rotor blades.

This roughness analysis was performed following ASTM D4417, a global standard for measuring the surface profile of blast-cleaned steel. A combination of methods Surface Comparator, Electronic Needle Gauging, and Replica Tape were used to match the curvatures of the blading and ensure the highest possible accuracy.

Surface Comparator: - Comparator is a flat, steel plate which contains a number of reference surface profiles, formed on a corrosion resistant metal. These reference profiles are then compared to the surface. There were five reference profile classes.

- 0.5S70 = 4.9 Microns
- 1S70 = 21.2 Microns
- 2S70 = 25.8 Microns
- 3S70 = 27.7 Microns
- 4S70 = 47.5 Microns

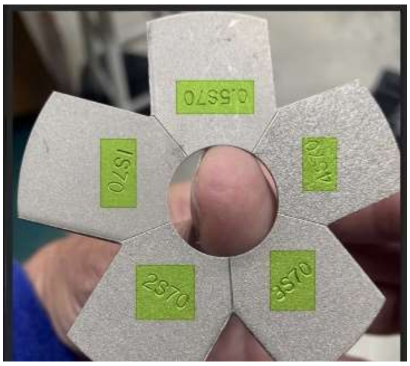

Figure 68 Surface Comparator

Electronic Needle Gauging: - Elcometer 224T Surface Profile Gauge was used as the needle gauge. Peak to valley surface profile heights were measured and recorded from this device. The tip angle was 60 degrees with conical shape. The device was calibrated using a glass plate and then used. Due to the shape of the impulse type of turbine buckets, this device was not used for the HP side.

Replica Tape method: - First the tape was placed on the surface and burnished it into the surface. Then the tape was removed and measured by using an electronic needle gauge. This method was used for the places where electronic needle gauges can't be used due to the shapes of the various buckets.

Figure 69 Elcometer 224T device

Figure 70 Replica Tapes

Results: -

The net value distribution over the high-pressure blades was around 30–40 microns, and the Surface Comparator showed similarity to segment 1S70. This profile remains the same for the face (In) and reverse (Out) regions, while the values similarly to 2S70 were showed by blades closer to the low-pressure segment on the outer/reverse side.

The low-pressure side was measured mainly using the electronic needle gauge. The pre-analysis was done by Elcometer 224T placing right over the uniform surfaces on the inner sides of blades. Max=60 µ; Min= -6.0 µ; Mean= 8.16 µ were the results and for the Outer/Reverse sides of blades Max=367.0 µ; Min= -7.0 µ; Mean= 90.42 µ were the results.

Therefore, there were minor rust or stains over the high-pressure segment while relatively greater observed at the low-pressure segment which is near to the condenser. The values taken from the surface comparator is available in **Annex 02.**

2.5.2.3 Steam Turbine Rotor Buckets Inspection.

Proper maintenance and inspection of turbine rotors and buckets primarily enhance efficiency. Turbines can suffer from water erosion, solid particle erosion, or chemical attacks during operation, which can degrade the quality of a new unit. By implementing procedures to monitor the unit's condition during operation and performing thorough inspections and maintenance during major outages, this can be significantly reduced. Consequently, an inspection was carried out to identify various types of defects in the rotor buckets. The following defects were the main focus of the inspection. The referenced drawings are available in **Annex 03**.

- Tenon erosion
- Cover grooving, rubbing, lifting, curling and mechanical damage.
- Tearing or thinning of vane edges.
- Pitting of grooving of vane body.
- Lifting of dovetails.
- Looseness of dovetail pins or keys.
- Erosion of vane section.
- Cracks in tie wire hole.

Results

The vane edges in every bucket of the 20th stage (both admission and exit) were eroded. The vane bodies were slightly eroded. The vane edges in every bucket of the 19th stage were also slightly eroded. There were no cracks in the tie wire holes. The dovetail pins and keys fitted very well, so there was no lifting of dovetails or looseness of dovetail pins or keys. There were small tenon erosions and mechanical damage. Pitting and grooving of the vane bodies were identified in the HP side stages 1 to 5.

3 Conclusion

My 26 weeks of industrial training at Yugadanavi CCPP was an invaluable opportunity that marked the beginning of my journey as a mechanical engineer. This training, facilitated by the Industrial Training Division of the University of Moratuwa in collaboration with NAITA, allowed me to apply the theoretical knowledge I had gained over the past five semesters in mechanical engineering. It also provided me with exposure to Sri Lanka's engineering and technical community, enabling me to learn from their experiences and knowledge while developing my interpersonal skills.

During this period, I had the privilege to work with modern technologies and gain hands-on experience with contemporary tools. I also gained insights into safety practices, project management, operational procedures, and the financial aspects of engineering projects. This experience helped me acquire essential skills such as teamwork, effective communication, time management, and problem-solving, which are crucial in the industry.

At Yugadanavi power plant, I learned extensively about mechanical maintenance, sand casting, and enhanced my skills in SolidWorks, Word, and Excel. I also understood the application of engineering technology and found solutions to various engineering challenges. I developed abilities in leadership, active listening, and making appropriate decisions through interactions with professionals in the field. Additionally, I had the rare opportunity to participate in major inspections of the steam turbine after 14 years, combustion inspections of the gas turbine, and hot gas path inspections, which provided me with invaluable insights.

Overall, this training period was immensely successful and a tremendous experience for me. I would like to express my heartfelt gratitude to everyone who contributed to my growth and development during this internship.

4 Annexes

Annex 01 (Failure types in bearings)

Figure 72 Fretting in CRP non drive end bearing

Figure 71 Wear in CRP non drive end bearing

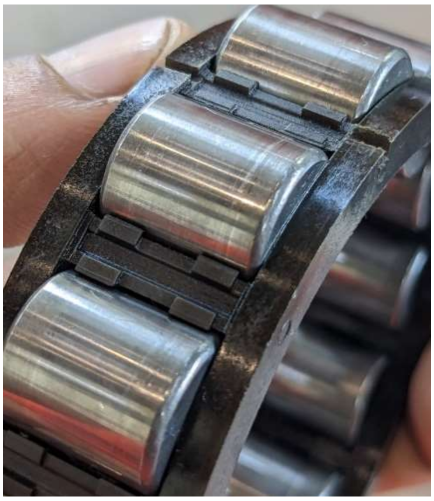

Figure 73 Scoring in CRP drive end bearing

Figure 74 Pitting in CRP drive end bearing

Figure 76 Wear in CT fan bearing

Figure 75 Movement of the running trace in CT fan bearing

Figure 78 Scratching in Water injection pump non drive end bearing

Figure 77 Cage Damage in Water injection pump non drive end bearing

Figure 79 Wear in Water injection pump Drive end bearing

Annex 02 (Steam Turbine Rotor blade roughness measurements using surface comparator)

Stage		Points														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Stage 1	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 2	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 3	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	2 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 4	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
Stage 5	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 6	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 7	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 8	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 9	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 10	In	2 S70	2 S70	2 S70	2 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	2 S70	2 S70	2 S70	2 S70	2 S70
	Out	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	1 S70	1 S70	1 S70	2 S70	2 S70	2 S70	2 S70	2 S70
Stage 11	In	2 S70	2 S70	2 S70	2 S70	1 S70	1 S70	1 S70	1 S70	1 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
	Out	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
Stage 12	In	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
	Out	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
Stage 13	In	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70
	Out	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
Stage 14	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
Stage 15	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70	2 S70
Stage 16	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
Stage 17	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70
Stage 18	In	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70	1 S70
	Out	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70
Stage 19	In	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70
	Out	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70
Stage 20	In	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70
	Out	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70	4 S70

Table 3 Steam Turbine Rotor blade roughness measurements using surface comparator

Annex 03

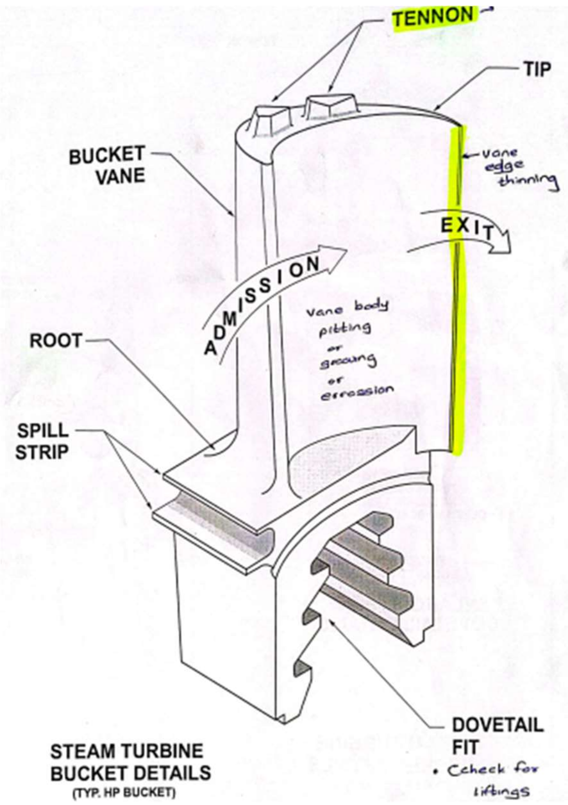

Figure 80 Steam Turbine HP side Bucket

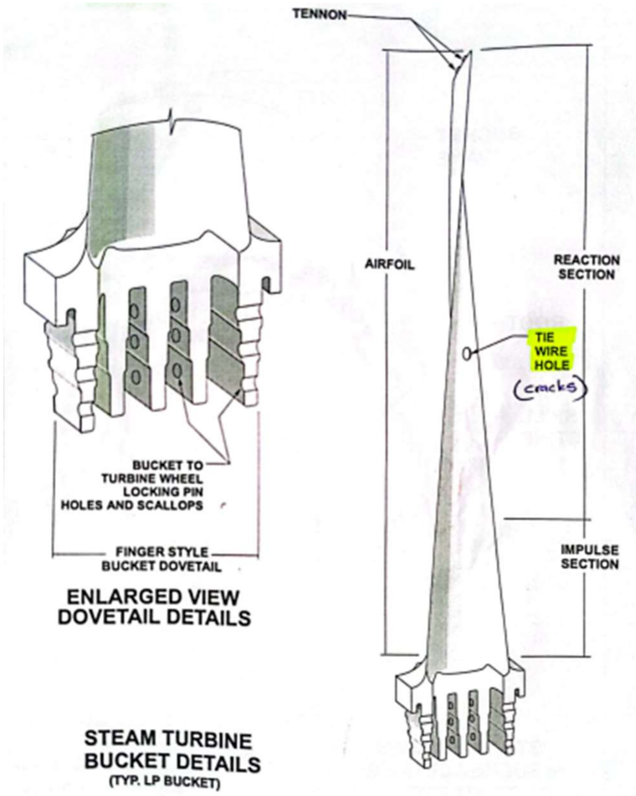

Figure 81 Steam Turbine LP side Bucket

5 References

(No date a) *Gas turbines | complete product range*. Available at: https://ftpturbine.com/DataBase/DBTemp/Images/downloads/Archive/04_Ebooks/08_GE-oil-and-gas-turbines-product-information .pdf (Accessed: 16 June 2024).

Catalogs (no date) *Industrial Steam Turbines - GE Steam Turbines - PDF Catalogs | Technical Documentation | Brochure*. Available at: https://pdf.directindustry.com/pdf/ge-steam-turbines/industrial-steam-turbines/116289-408113 .html (Accessed: 16 June 2024).

Ircengg (2024) *Liquid penetrant test: Pros, cons & other options - IRC engg.*, *IRC Engineering Services (India) Pvt. Ltd.* Available at: https://ircengg.com/liquid-penetrant-testing/ (Accessed: 16 June 2024).

High quality, Reliable Delivery & competitively priced (no date) *Gas Turbine Parts & Services*. Available at: https://gtps-inc.com/works/product/ (Accessed: 16 June 2024).

XT660 - Advanced Shaft Alignment system (no date) *Easy*. Available at: https://easylaser.com/en-us/products/shaft-alignment/xt660-dot-laser (Accessed: 16 June 2024).

38DL plus (no date) *38DL PLUS Ultrasonic Thickness Gage*. Available at: https://www.olympus-ims.com/en/38dl-plus/ (Accessed: 16 June 2024).

Yugadanavi Power Station (2023) *Wikipedia*. Available at: https://en.wikipedia.org/wiki/Yugadanavi_Power_Station (Accessed: 16 June 2024).